

DIKSHA PHULORIA

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SUMMARY

Data Science graduate from Rutgers University (GPA 3.9). I've always seen building solutions as solving a carefully placed puzzle and that curiosity has taken me from enterprise analytics to agentic AI systems and mechanistic interpretability research.

EDUCATION

Rutgers University

M. S. Data Science - GPA: 3.9/4

August 2024 - May 2026

New Brunswick, NJ, USA

Manipal University Jaipur

B. Tech, Information Technology - GPA: 8.07/10

July 2017 - June 2021

Rajasthan, India

WORK EXPERIENCE

Accenture

Analyst, Custom Software Engineering

May 2021 - June 2023

Gurugram, India

- Analyzed and visualized financial data for a banking client using Tableau and Power BI, building dashboards that improved reporting reliability and supported data-driven decision making across business units.
- Investigated and resolved 200+ data pipeline and application issues via ServiceNow, identifying root causes in data workflows and ensuring data integrity across critical banking systems.

PROJECTS

NutriSync - Agentic AI Meal Planning System

February 2026 - May 2026

Next.js 15, FastAPI, PostgreSQL, Google Gemini, Docker, TypeScript, Tailwind CSS

- Architected a full-stack agentic AI platform unifying meal planning, nutrition tracking, pantry management, and grocery shopping with Google Gemini, using Human-in-the-Loop (HITL) checkpoints at every decision stage for reliability in a health-critical context.
- Engineered cross-module data propagation where selecting a meal auto-deducts pantry inventory, recalculates macros, and refreshes the grocery list; backed by a 12-table PostgreSQL schema enriched from Spoonacular and USDA APIs.

Interpreting Climate Representations in GPT-2 via Sparse Autoencoders

February 2026 - May 2026

Python, PyTorch, Hugging Face Transformers, Google Colab (A100 GPU)

- Trained a 32,768-feature SAE on 839,773 GPT-2 Medium layer-17 activations, extending Anthropic's mechanistic interpretability framework to the climate domain for the first time.
- Discovered 21 climate features across four selectivity tiers; Carbon (15.96×) and Greenhouse/Emissions (8.56×) were most selective, while Glaciers and Atmosphere proved non-climate-selective, demonstrating surface-level token inspection is insufficient.

Carbon Emission Tracking in Language Model Fine-Tuning

Sep 2025 – Dec 2025

Python, Hugging Face, CodeCarbon, SQuAD v2.0

- Benchmarked DistilBERT, BERT, RoBERTa, and GPT-2 across full fine-tuning, LoRA, and few-shot learning using CodeCarbon to track CO₂ emissions per training run.
- RoBERTa + LoRA was optimal (F1=0.7426, 9.56 F1/kg CO₂), achieving ~95% performance with 12–24% lower emissions; introduced F1-per-kg-CO₂ as a replicable sustainability metric.

NYC 311 Service Requests Analytics Pipeline

Microsoft Fabric, Azure Data Factory, OneLake, PySpark, SQL, Power BI, Python

- Built an end-to-end data pipeline using Microsoft Fabric (Data Factory, OneLake, Notebooks) to ingest, clean, and analyze 1M+ NYC 311 service requests, surfacing borough-level trends via a Power BI dashboard.

SKILLS

Languages and Querying: Python, SQL, R

Machine Learning and AI: Predictive Modeling, NLP, AI and LLMs, Causal Inference, Feature Engineering

Data Analysis and Visualization: Tableau, Power BI, Pandas, Exploratory Data Analysis (EDA), Data Wrangling

Databases and Platforms: PostgreSQL, MySQL, Snowflake, Microsoft Fabric, Azure Data Factory (ADF), OneLake, Streamlit, Google Colab, ServiceNow, Git, VS Code

CERTIFICATIONS

[Artificial Intelligence Fundamentals - IBM](#)

May 2026

[SnowPro Core – Snowflake](#)

Aug 2025 – Aug 2027